1.List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Here are three real-life scenarios where a **multimeter** is used to troubleshoot a computer or smartphone:

Scenario	**How the Multimeter Helps (One Line)**
1. Computer does not power on	Measures the power supply voltage to check if the PSU is providing the correct output.
2. Smartphone battery not charging	Tests the battery and charging port voltage to determine if the battery or charger is faulty.
3. Laptop charging problem	Checks the laptop charger and DC power jack voltage to identify a damaged adapter or charging circuit.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

- After watching a typical POST card demonstration, I observed that the POST card displays   as the BIOS checks different hardware components during startup. The codes change as each test is completed, and if the system stops on a particular code, it indicates the hardware component causing the boot failure. In the demo, the POST card helped identify where the startup process stopped, allowing the faulty hardware (such as memory, CPU, or motherboard) to be diagnosed without the computer successfully booting.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

1. Download any one of these free diagnostic tools:

- HWMonitor
- Speccy
- CrystalDiskInfo

2. Install the software and open it.

3. Wait for the main dashboard to load. It should display information such as:

- CPU temperature
- RAM details
- Storage/SSD or HDD health
- Motherboard information
- System temperatures and voltages (depending on the tool)

4.Take a screenshot of the main dashboard:

- Press **Windows + Shift + S** (Snipping Tool) or **PrtScn**.
- Save the image.

5.Before submitting:

- Blur or crop out your **computer name**, **Windows username**, **serial number**, or any other personal information if visible.

6.Submit the screenshot as your assignment.

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	**What it Checks**	**Example Problem Detected**
Multimeter	Measures electrical values such as voltage, current, and resistance in hardware components.	Faulty power supply (PSU), dead laptop charger, or weak CMOS battery.
POST Card	Displays POST (Power-On Self-Test) error codes during the computer's startup process.	Faulty RAM, CPU, or motherboard causing the computer to fail to boot.
Software Diagnostics	Monitors system health, hardware status, temperatures, storage health, and performance.	Overheating CPU, failing hard drive/SSD, or low available memory affecting system performance.